



Web Digital Vertical Obstruction File (WebDVOF) Version Description Document

Foundation Sustainment Services (FS2)

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Revision History

Version #	Implemented By	Revision Date	Reason
6.0.1	Christi Curtis	03/04/2025	Initial release of this version of the product.

Authorization Notification

Any authorization applies only to the current version of the document.

Role	Name	Company	Job Title
Prepared By			WebDVOF Product Owner
Reviewed By			CM Lead / Test Engineer
Checked By			Configuration Manager
Authorized By			Aero Program Manager

Approval

The undersigned have read this VDD and approve its contents.

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Date

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Date



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1 Scope

1.1 Introduction

This document is the *Version Description Document (VDD) for Air WebDVOF Version 6.0.1*

This VDD details the software components to be delivered with Air WebDVOF Version 6.0.1. This document describes the features included in the delivery of the software build, at entrance to test. It includes the build artifacts for deployment, any known issues with the delivery, and the changes included in the software version.

1.2 System Overview

WebDVOF is a web-based application that supports production and dissemination of aeronautical products containing vertical obstruction data. The system allows end users to download predefined files and products generated ahead of time and allows end users to submit custom queries and generate files and products with the query results in real time. Various file formats are supported by the system.

The system consists of multiple components: a web application providing a user interface via a web browser, a server component providing a RESTful API and implementing business logic, a processor component that performs data queries and generates files/products, a persistent volume/file storage component where generated or uploaded files are stored for use by the system, and a relational database with a spatial component providing data storage and querying capability. In addition, the system relies on external services such as an OAuth 2 Authorization Server (for user authentication) and a Map Tile Server (for query preview image maps) to provide some capabilities.

The system is designed for deployment on the JWICS, SIPRNet, and NIPRNet networks, and shall have fail-over capability. It is designed to run in a Platform-as-a-Service (PaaS) cloud environment such as OpenShift.

1.3 Document Overview

The Air WebDVOF VDD provides the list of software delivered for Version 6.0.1 of the product and the capabilities of that software. This document details a single release of the product and will be updated only to correct information specific to version 6.0.1 of the product. Follow-on versions of the product will be documented in a separate VDD specific to that version of the product.

The document is organized as follows:

- Section 1 – Scope
- Section 2 – Version Description



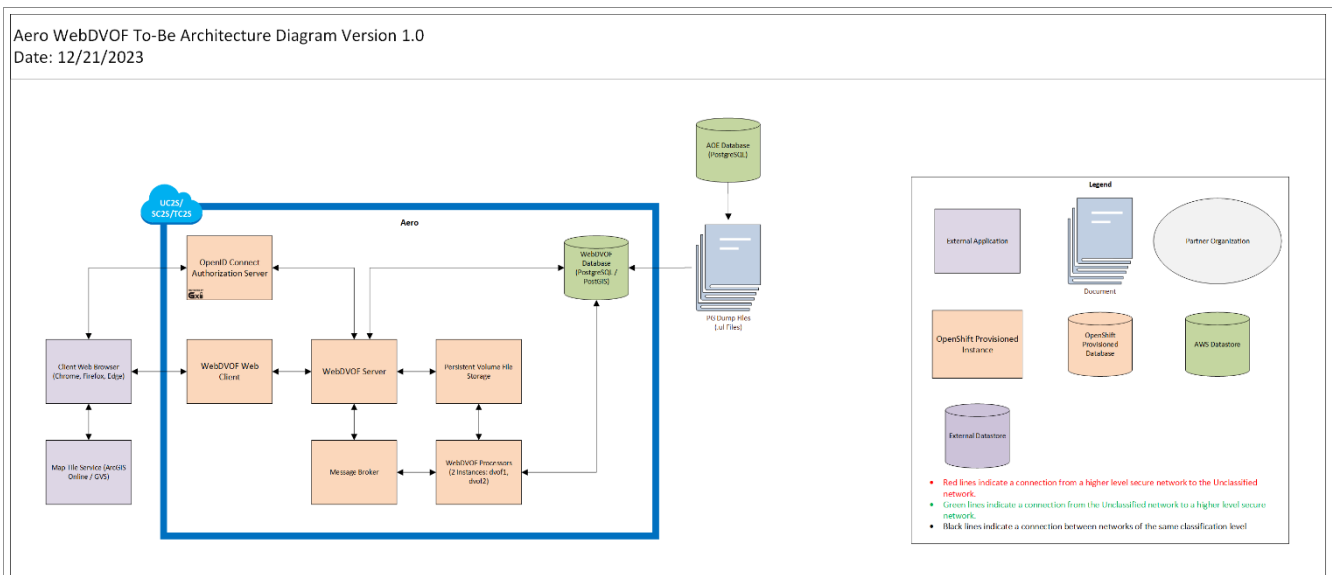
2 Version Description

This release represents an update to software and database used by the WebDVOF application. The application has been updated with several improvements and bug fixes based on customer feedback, including updates to Predefined Query management, Shapefile file formatting, and Administrator Data Management. To ensure that the RabbitMQ software that WebDVOF depends on can be updated and maintained, the team created its own Docker image for RabbitMQ to replace the existing RabbitMQ Docker image originally sourced from SLO. Additionally, this release includes an upgrade of the database to PostgreSQL 16.3 and PostGIS 3.4.1. Lastly, several other dependency libraries were updated to reduce technical debt and to maintain security posture.

2.1 Deployment Environment

The system will be deployed to the NGA KUBIC environment, utilizing the OpenShift platform. Relational databases will be deployed using AWS RDS instances linked to and managed by the OpenShift platform, and Persistent Volume Service and OAuth 2 Authorization services will be provided by the OpenShift platform. The WebDVOF application components (server, processor, and web client) will be deployed to OpenShift pods using containers defined by the application team. These components will be scalable based on demand, so additional instances of the application components can be started up and used dynamically on an as-needed basis, in order to maintain or improve system performance. The OpenShift platform will be responsible for load balancing, logging aggregation, and external URL mapping.

Architecture Diagram





2.2 Software Functionality and Capabilities

The WebDVOF system will enable users to query vertical obstruction data and generate custom files/products containing the query results. The user will be able to select from various formatting and filtering options, including spatial filters, when generating custom files/products. Users will be able to save these queries and edit or re-execute them at a later time.

In addition, users will be able to download predefined query files/products from a dedicated download page. Administrator users will be able to generate predefined query files/products and manage which files appear in the predefined query download page.

Administrator users will also be able to initiate and monitor data loads into an offline database schema, while normal users are still able to query and access data in a separate online database schema. Administrator users will be able to publish a data load once it is complete, so that it becomes the active schema accessed by normal users.

Lastly, Administrator users will also be able to monitor query logs and manage system settings including Notices to Users, which will appear in the application user interface.

2.3 Inventory of Software Components

This section documents the software configuration for WebDVOF.

2.3.1 Software Deliverables

WebDVOF will deliver custom software packages and build artifacts containing the application components, including but not limited to Java "jar" files and JavaScript, HTML, and CSS scripts. These packages will include code library dependencies along with custom code to ensure that the application has all of the software needed to execute.

2.3.2 Software List

The WebDVOF Version 6.0.1 Software Lists for COTS and Open Source code can be found in Table 2.3.2.

Table 2.3.2 - Software List

Software	Version	SWID	Vendor	Control #	Notes
Web Digital Vertical Obstruction File (WebDVOF)	6.0.1	31987	Aeronautical Navigation Office (SFA)	202502280018H	
WebDVOF Server					
Java JDK 21 (from CORE image)	21	30138	Oracle Corporation	20240206006	
Spring Boot	3.4.1	31248	VMWare Tanzu Labs	202501160083H	
io.spring.dependency-management	1.1.7	31348	Gradle	202501160088H	
spring-boot-starter-actuator	3.4.1	20386	VMWare Tanzu Labs	202501160078H	
spring-boot-starter-amqp	3.4.1	23134	VMWare Tanzu Labs	202501140083H	
spring-boot-starter-data-jpa	3.4.1	19033	VMWare Tanzu Labs	202501160077H	
spring-boot-starter-json	3.4.1	21035	VMWare Tanzu Labs	202501160075H	



spring-boot-starter-oauth2-resource-server	3.4.1	26348	VMWare Tanzu Labs	202501160074H	
spring-boot-starter-validation	3.4.1	21642	VMWare Tanzu Labs	202501150124H	
spring-boot-starter-web	3.4.1	19080	VMWare Tanzu Labs	202501150119H	
flyway-core	10.20.1	25944	Redgate Software. LTD	202501160107H	
flyway-database-postgresql	10.20.1	31932	Redgate Software. LTD	202501160106H	
hibernate-spatial	6.6.4.Final	6488	Hibernate	202501160101H	
jdom2	2.0.6.1	26167	Jason Hunter	20221206026	
PostgreSQL JDBC Driver	42.7.4	7440	The PostgreSQL Global Dev Group	20241007071	
commons-codec	1.17.1	25499	Apache Commons	202501150088H	
jaxen	2.0.0	6435	Elliot Rusty Harold	20240521136	
springdoc-openapi-starter-webmvc-ui	2.7.0	27469	SpringDoc	202501150108H	
jackson-datatype-jts	1.2.10	14390	Bedatadriven	20230309057	
commons-io	2.18.0	23683	Apache Software Foundation	202501150083H	
WebDVOF Processor					
Java JDK 21 (from CORE image)	21	30138	Oracle Corporation	20240206006	
Spring Boot	3.4.1	31248	VMWare Tanzu Labs	202501160083H	
io.spring.dependency-management	1.1.7	31348	Gradle	202501160088H	
spring-boot-starter-amqp	3.4.1	23134	VMWare Tanzu Labs	202501140083H	
spring-boot-starter-data-jpa	3.4.1	19033	VMWare Tanzu Labs	202501160077H	
commons-lang3	3.17.0	12614	Apache Software Foundation	20231113022	
hibernate-spatial	6.6.4.Final	6488	Hibernate	20240501056	
commons-codec	1.17.1	25499	Apache Commons	202501150088H	
PostgreSQL JDBC Driver	42.7.4	7440	The PostgreSQL Global Dev Group	20241007071	
WebDVOF Web Client					
Node.js 20 (from CORE image)	20.18.2	28476	OpenJS Foundation	202501240091H	
npm (from CORE image)	10.9.2	16127	Isaac Z. Schlueter	202501140041H	
@emotion/react	11.14.0	24125	Mateusz Burzynski	20241210005	
@emotion/styled	11.14.0	31193	Emma Hamilton	202501150103H	
@fontsource/roboto	5.1.1	32419	fontsource	202501150102H	
@hookform/resolvers	3.9.1	24224	Bill Luo	20241029104	
@mui/icons-material	6.3.0	24615	Material-UI	202501150100H	
@mui/material	6.3.0	24255	Oliver Tassinari	202501150099H	
axios	1.7.9	7599	Jason Saayman	202501150089H	
luxon	3.5.0	18564	Isaac Cambron	20240923125	
next.js	15.1.3	23397	Vercel	202501150045H	
next-auth.js	4.24.11	26762	Ian Collins	202501150039H	
ol (OpenLayers)	10.3.1	29467	OpenLayers	202501140092H	
react	19.0.0	31202	facebook/react	20241219013	
react-dom	19.0.0	31097	Joseph Savona	20241219010	
react-hook-form	7.54.2	20912	Bill Luo	202501140089H	
react-number-format	5.4.3	21782	Sudhanshu Yadav	202501140086H	
swr	2.3.0	22198	Vercel	202501140081H	
zod	3.24.1	27418	Colin McDonnel	202501140076H	
RabbitMQ - container					
UBI8 (CORE Container)	8.10	27525	National Geospatial-Intelligence Agency (NGA)	202501140060H	
Erlang	26.2.5.3	5018	Ericsson Computer Science Laboratory	20241017062	



rabbitMQ	4.0.2	29250	Iron Bank	20241017061	
Database					
Postgresql Server	16.6-3.4-postgis	29582	EnterpriseDB	20241220016	
WebDVOF development dependencies					
WebDVOF Web Client – development dependencies					
@testing-library/react	16.1.0	21140	Kent C. Dodds	202501150096H	
@types/luxon	3.4.2	29075	Matt R. Wilson	20240415022	
@types/node	22.10.2	13320	Nathan Shively-Sanders	202501150095H	
@types/react	19.0.2	18058	Nathan Shively-Sanders	202501150094H	
@types/react-dom	19.0.2	18053	Nathan Shively-Sanders	20241219011	
@typescript-eslint/eslint-plugin	8.19.0	30300	Eclipsesource	202501150093H	
@typescript-eslint/parser	8.19.0	19970	Josh Goldberg	202501150092H	
@vitejs/plugin-react	4.3.4	30081	Vite	202501150091H	
@vitejs/coverage-v8	2.1.8	33249	Vladimir Rutsky	202501150090H	
eslint	9.17.0	15393	Nicholas Zakas	202501150061H	
eslint-config-next	15.1.3	25781	Vercel	202501150059H	
eslint-config-prettier	9.1.0	17083	Simon Lydell	20240102035	
eslint-plugin-prettier	5.2.1	17082	Ben Scott	20241002034	
eslint-plugin-testing-library	7.1.1	30754	Mario Beltran	202501150058H	
jsdom	25.0.1	16483	Domenic Denicola	20241003047	
msw	2.7.0	21609	kettanaito	202501150055H	
prettier	3.4.2	26610	Fisker Cheung	202501140090H	
typescript	5.7.2	12767	Microsoft	20241203019	
typescript-eslint	8.19.0	18956	James Henry	202501140079H	
vitest	2.1.8	30678	Vladimir	202501140078H	
WebDVOF Server and Processor development dependencies					
Gradle	8.12	12465	Gradle	202501160104H	
spring-rabbit-test	3.2.1	29428	VMWare Tanzu Labs	202501140082H	
spring-boot-starter-test	3.4.1	19039	VMWare Tanzu Labs	202501160069H	
junit-platform-launcher	1.11.4	31215	Junit	202501150056H	

2.4 Changes Installed

As was noted in Section 1.3, this is considered to be in initial release of the 6.0.1 WebDVOF product so changes from the previous version are not recorded in this VDD.

Table 2.4 includes defects recorded in testing as well as any other internally discovered defects that were resolved.

- Issue Types:
- Test Problem Report (TPR): DRs written by the NTO during testing
- System Problem Report (SPR): DRs written by problems found by the PMO, customer, or external systems.



Table 2.4 - Changes Installed

PBI/INC #	Key	Issue Type	Summary

2.5 Installation Instructions

The application components will be installed using the NGA CORE services GitLab CI/CD pipeline. No special install instructions will be needed once the pipeline is set up and configured.

(Instructions on how to manually install and configure the application, if needed for this document, are To Be Determined).

2.6 Possible Problems and Known Errors

Table 2.6 Version 6.0.1 Possible Problems and Known Errors

Issue ID	Summary	Description